

3

5

LCM AND HCF

WITH MULTIPLES

7

2

What are Multiples?

Multiples are the numbers that you get from multiplying a number by another number.

2

$$2 \times 1 = 2$$

$$2 \times 2 = 4$$

$$2 \times 3 = 6$$

$$2 \times 4 = 8$$

$$2 \times 5 = 10$$

**The multiples
of 2 are:**

2, 4, 6, 8, 10
and so on...

Note that the smallest
multiple of any number
is the number itself.

KEYWORDS

HCF

HCF stands for the highest common factor of two numbers

LCM

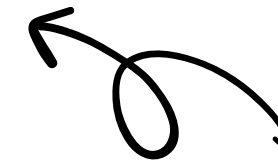
LCM stands for the lowest common multiple of two numbers.

LOWEST COMMON MULTIPLE

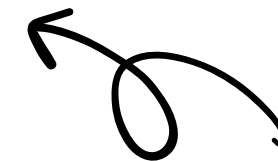
Find the LCM of 12 and 16

Multiples appear
in the number's
times tables

Multiples of 12: 12, 24, 36, 48, 60



Multiples of 16: 16, 32, 48, 64, 80



LCM = 48

HIGHEST COMMON FACTOR

Find the HCF of 12 and 16

Factors are
numbers that
divide into a
number exactly

Factors of 12: 1, 2, 3, 4, 6, 12



Factors of 16: 1, 2, 4, 8, 16



HCF = 4

USING PRIME FACTORISATION

Prime factorisation can be used to find the LCM and HCF of large numbers using the following steps:

1

Write each number
as a product of its
prime factors

2

Place the prime
factors of both
numbers in a
venn diagram

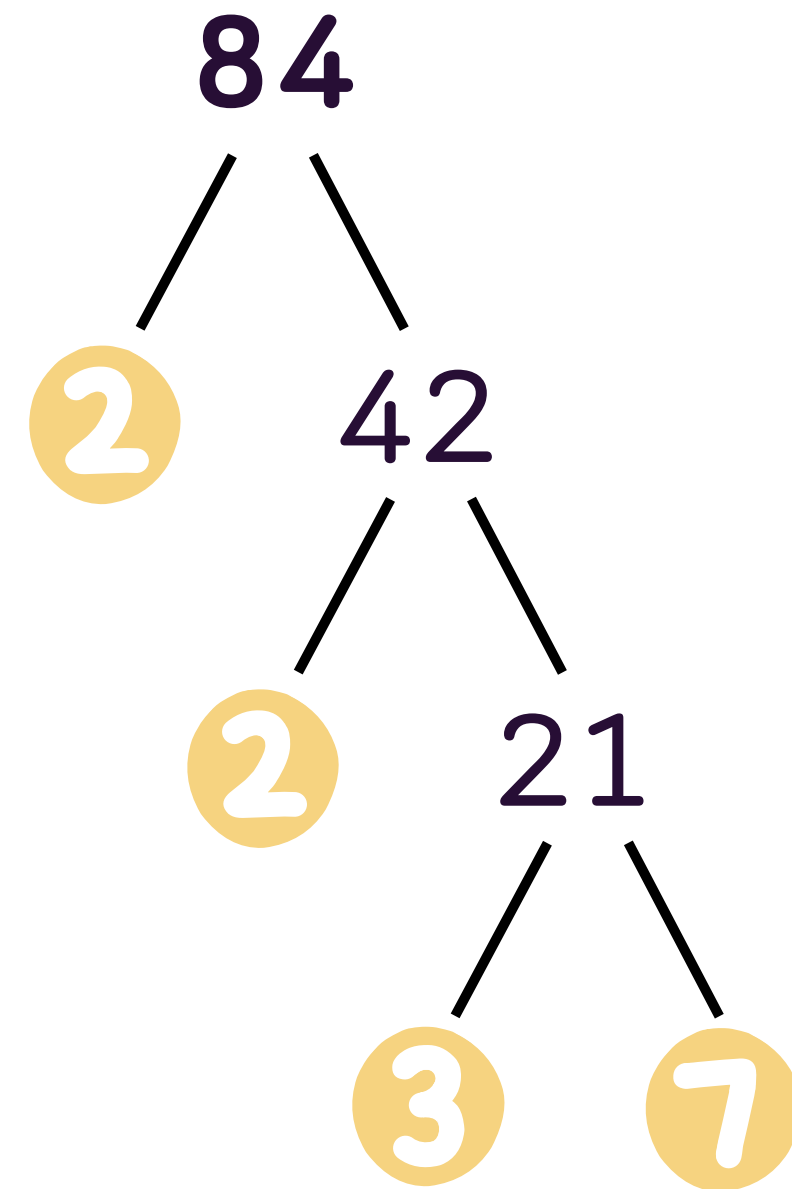
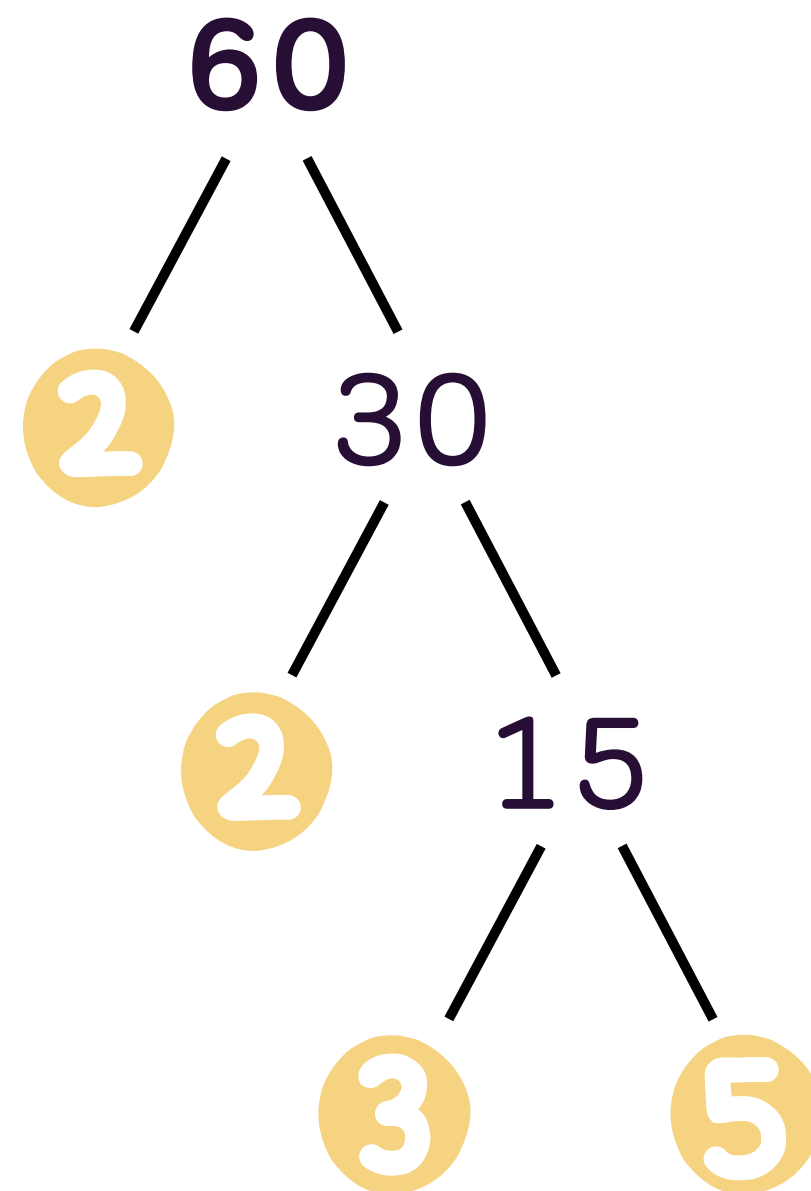
3

Calculate the LCM
and HCF from the
venn diagram

1

FIND THE LCM AND HCF OF 60 AND 84

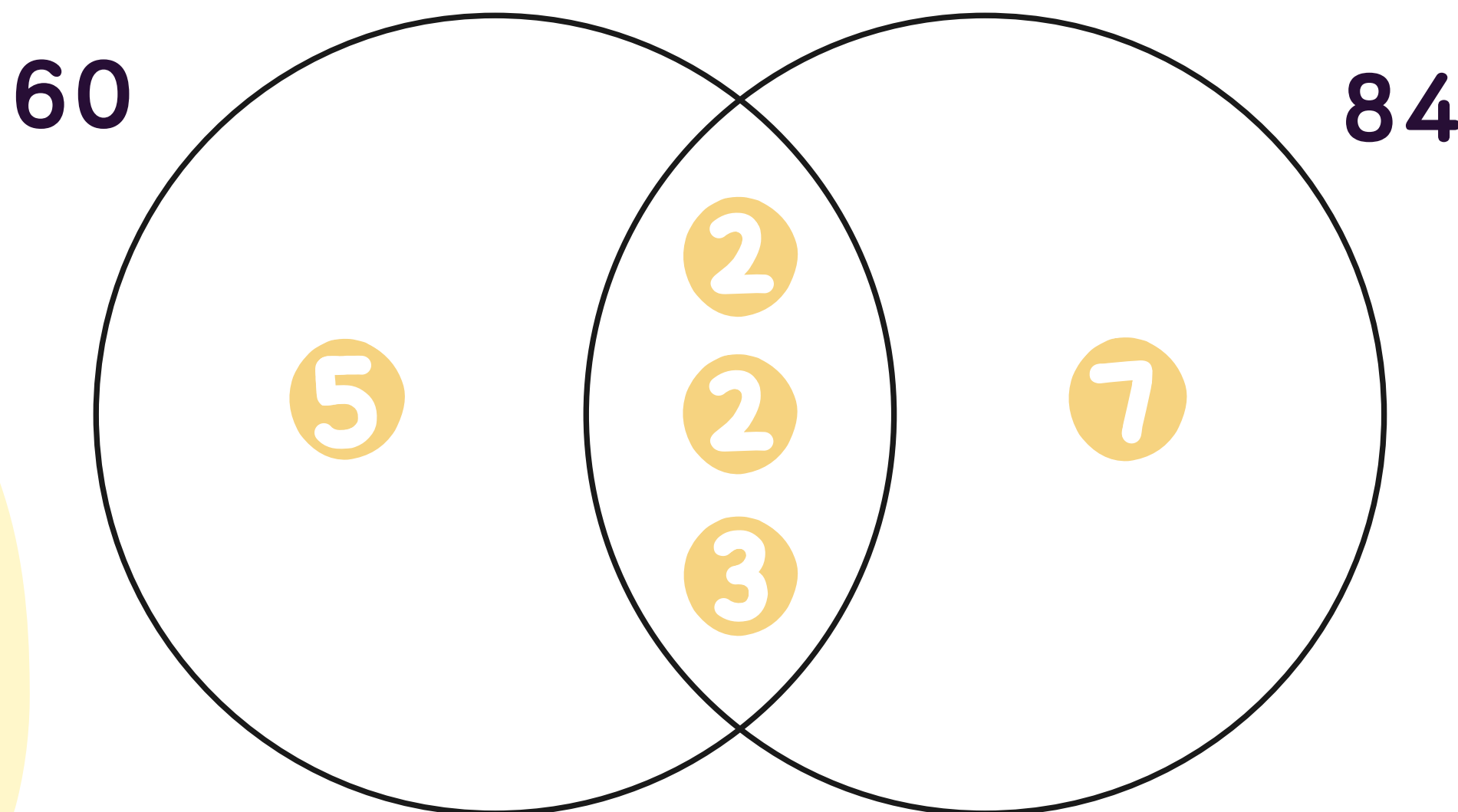
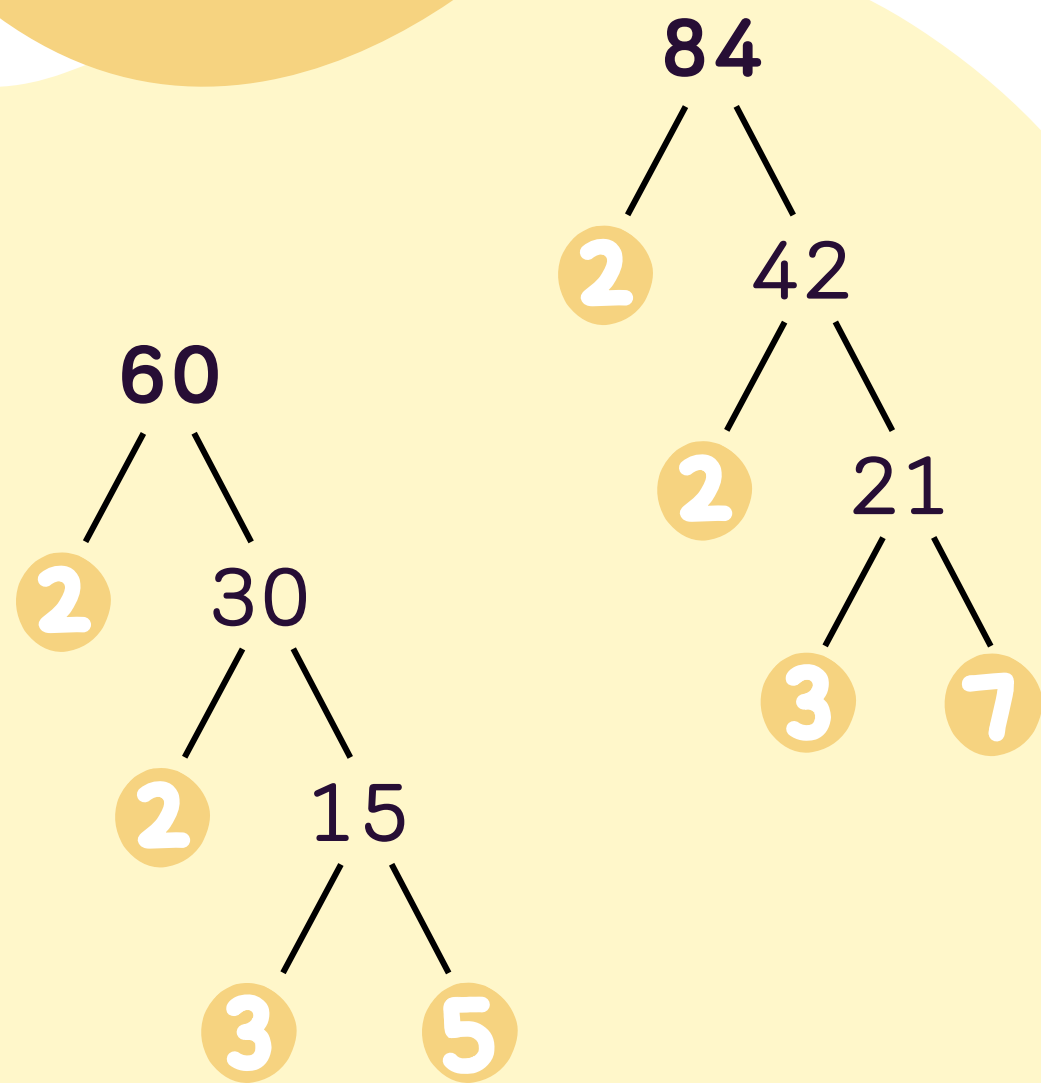
Write each number as a product of its prime factors



2

FIND THE LCM AND HCF OF 60 AND 84

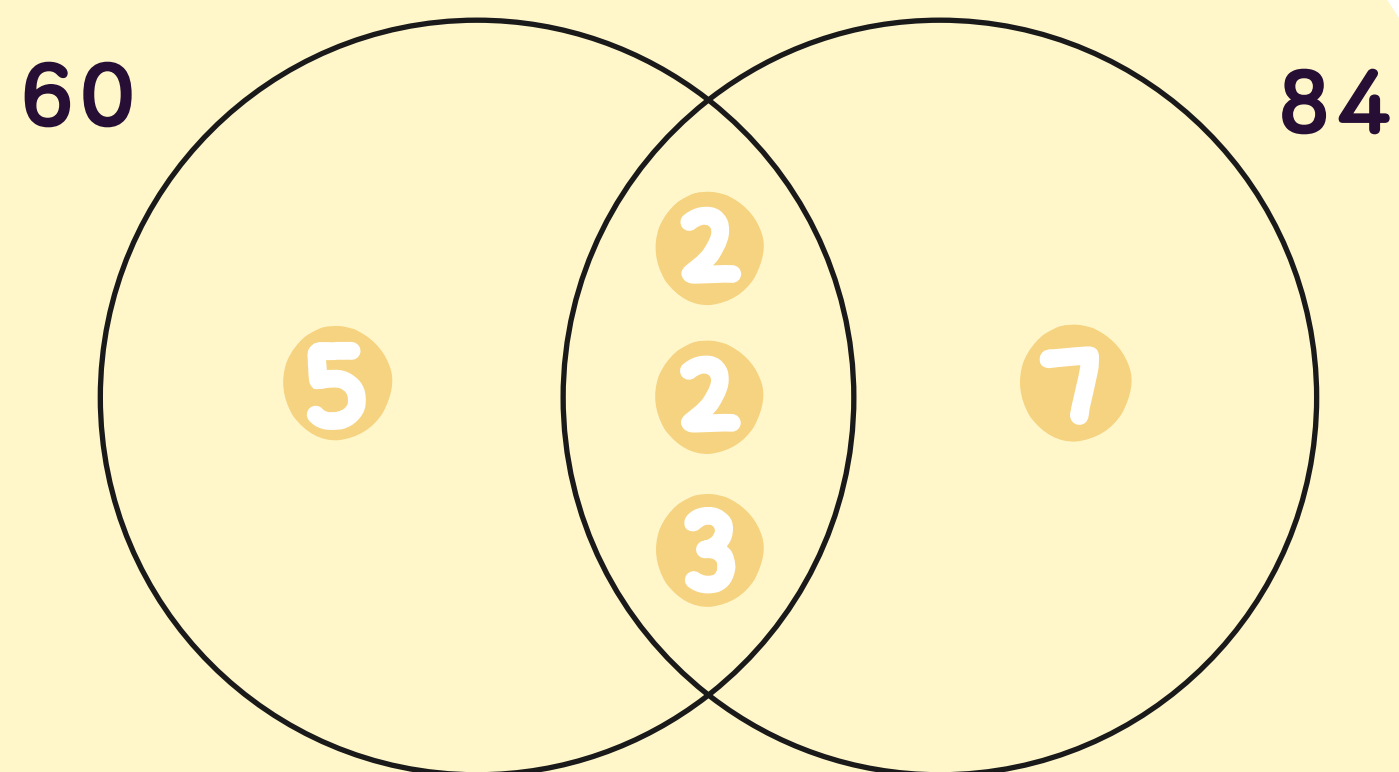
Place the prime factors of both numbers in a venn diagram



3

FIND THE LCM AND HCF OF 60 AND 84

Calculate the LCM and HCF from the venn diagram



$$\text{LCM} = 2 \times 2 \times 3 \times 5 \times 7 = 420$$

All the numbers in the venn diagram multiplied together

$$\text{HCF} = 2 \times 2 \times 3 = 12$$

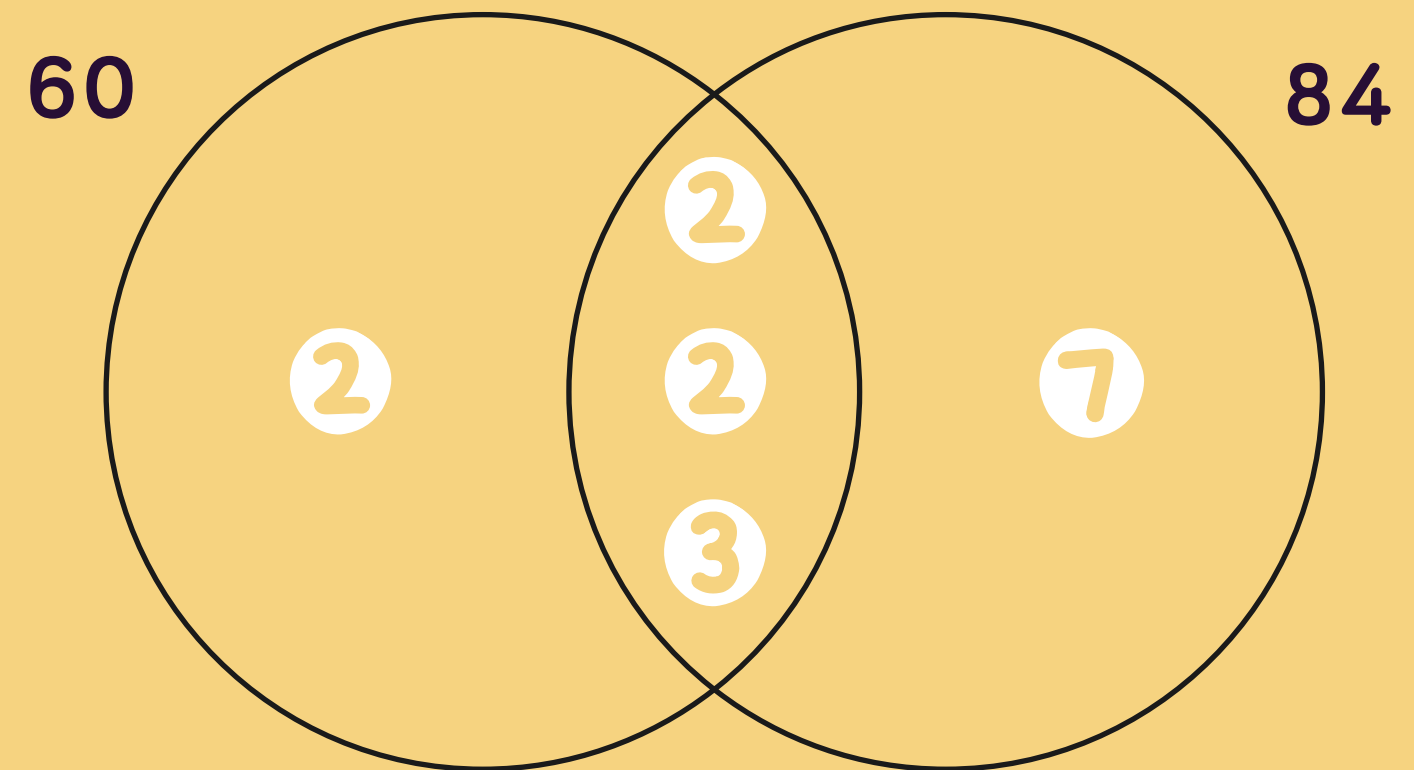
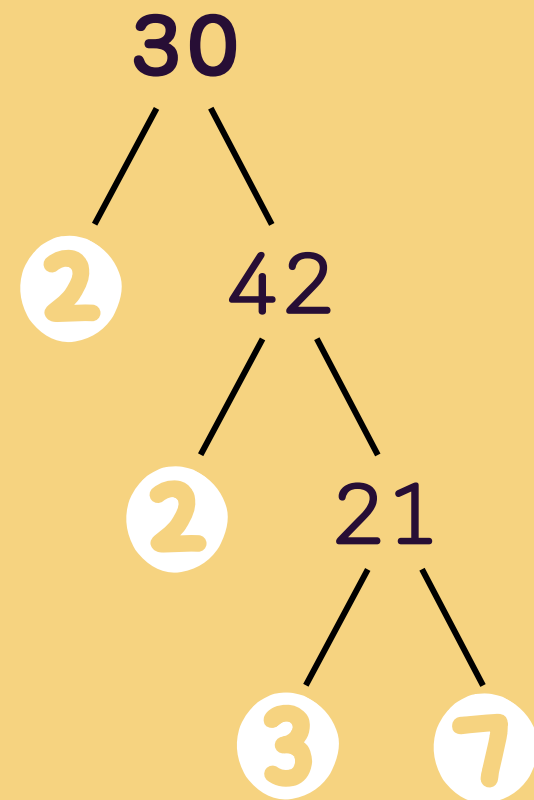
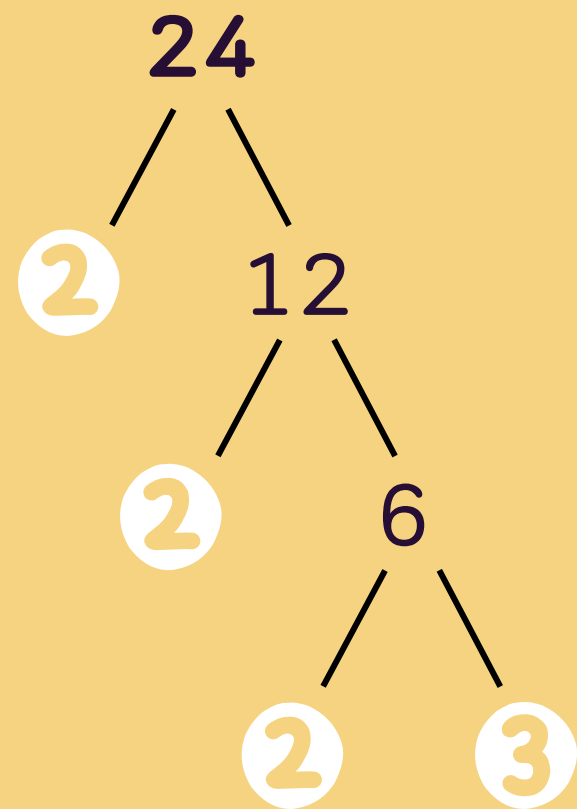
The numbers in the intersection of the venn diagram are multiplied together

YOUR TURN

**FIND THE LCM AND HCF OF
24 AND 30**

SOLUTION:

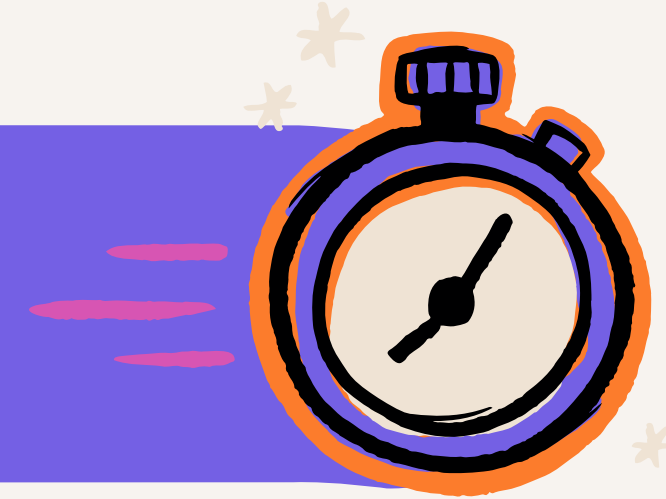
FIND THE LCM AND HCF OF 24 AND 30



$$\text{LCM} = 2 \times 2 \times 3 \times 5 \times 7 = 420$$

$$\text{HCF} = 2 \times 2 \times 3 = 12$$

Practice Questions



LCM

Find the lowest common multiple:

1. 8 and 15

2. 9 and 14

3. 12 and 21

HCF

Find the highest common factor:

1. 16 and 34

2. 36 and 60

3. 56 and 98

Answer Key



LCM

Find the lowest common multiple:

1. 8 and 15 $\text{LCM} = 120$

2. 9 and 14 $\text{LCM} = 126$

3. 12 and 21 $\text{LCM} = 63$

HCF

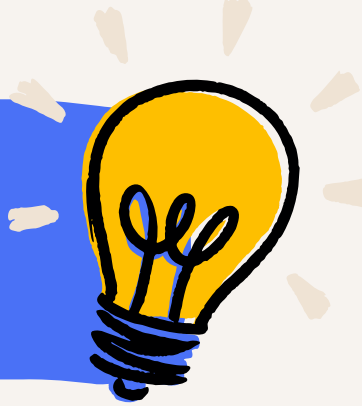
Find the highest common factor:

1. 16 and 34 $\text{HCF} = 2$

2. 36 and 60 $\text{HCF} = 12$

3. 56 and 98 $\text{HCF} = 14$

Word Problems



1.

A park has two different light displays. One display lights up every 12 minutes, and the other lights up every 15 minutes. If both displays light up at 6:00 PM, when will they next light up at the same time?

2.

Hannah has two pieces of card. The longer piece of card is 28cm and the shorter piece is 20cm. She wants to cut them into the smallest number of equal sized pieces. What is the size of each piece she should cut?



Solutions



1.

A park has two different light displays. One display lights up every 12 minutes, and the other lights up every 15 minutes. If both displays light up at 6:00 PM, when will they next light up at the same time?

Multiples of 12: 12, 24, 36, 48, 60

Multiples of 15: 15, 30, 45, 60

LCM = 7:00 PM

2.

Hannah has two pieces of card. The longer piece of card is 28cm and the shorter piece is 20cm. She wants to cut them into the smallest number of equal sized pieces. What is the size of each piece she should cut?

Factors of 28: 1, 2, 4, 7, 14, 28

Factors of 20: 1, 2, 4, 5, 10, 20

HCF = 4 cm (12 pieces)





SQUARE NUMBERS

Square number = a whole number that you can get by multiplying another whole number by itself.

Examples:

- 49 is a square number because it is the product of 7×7 .
- 100 is a square number because it is the product of 10×10 .



Try this!

Write the first 8 multiples of 18. Which of them are also multiples of 6?

A school bell rings every 9 minutes. Another bell rings every 12 minutes. If both rang together at 9:00 am, what time will they ring together again?

What are the first 5 multiples of 10?

Find the smallest common multiple of 12, 15, and 20.

Find the smallest number greater than 100 that is a multiple of both 8 and 15.

Try this!

First 8 multiples of
18: 18, 36, 54, 72,
90, 108, 126, 144.

All of these are
multiples of 6.

Bells ring every 9 and 12 min
→ LCM = 36 min → Next
together at 9:36 am

10, 20,
30, 40,
50

LCM of 12, 15, 20 →
 $12 = 2^2 \times 3$, $15 = 3 \times 5$,
 $20 = 2^2 \times 5$ → LCM =
60

Multiples of 8 and
15 → LCM =
 $2^3 \times 3 \times 5 = 120$ →
Smallest >100 is
120

Well Done!

Thank you!

